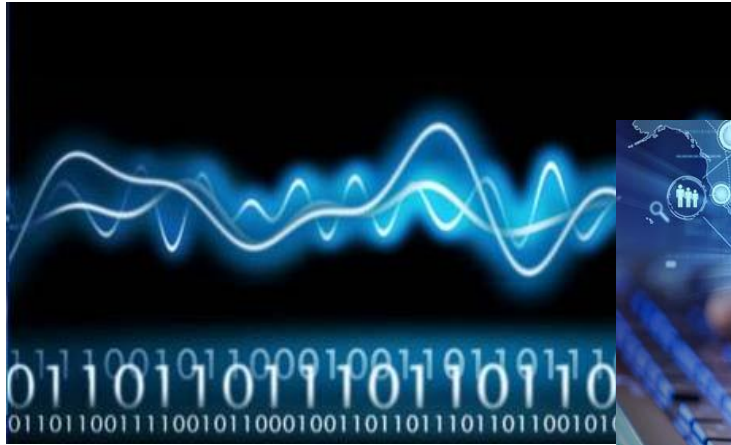


EEE/ECE 341: Introduction to Communication Engineering



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Introduction to Communication Systems

Multitude of Communications

Communication: Transfer of information

- ☐ Telephone network
- ☐ Internet
- ☐ Radio and TV broadcast
- ☐ Wireless mobile communications
- ☐ Wi-Fi
- ☐ Satellite communications
- ☐ Deep-space communications
- ☐ Smart power grid, healthcare...

- ☐ **Analogue communications**

- AM, FM

- ☐ **Digital communications**

- Transfer of information in digits
- Dominant technology today
- Broadband, DSL, ADSL, 3G, 4G



History of Communication: Early

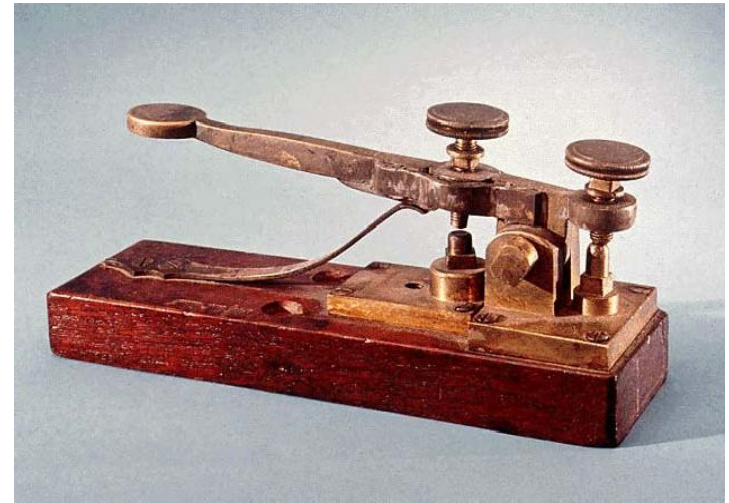
- ❑ Before human beings created languages and alphabets, they communicated with **both sound and body language**
- ❑ They have used **smoke and light** for communication
- ❑ **Drawing** is also used for communication at the early stage



History of Communication: Developing Technology

❑ 1799: Alessandro Volta
invented electric battery

❑ 1831: Professor Joseph Henry,
Albany, NY – Ring a bell at a
distance by connecting and
disconnecting wires



History of Communication

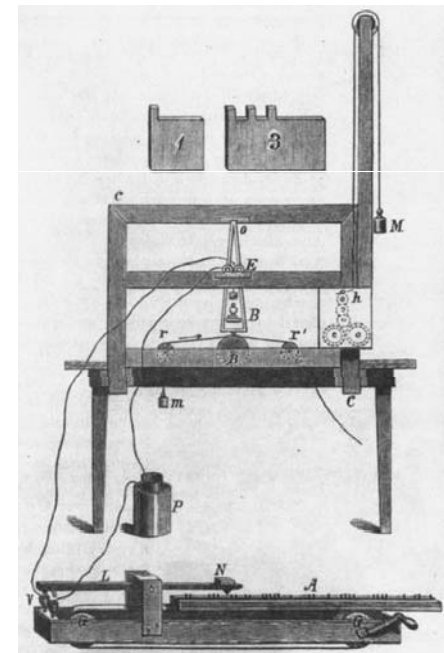
❑ 1837: Samuel Morse demonstrated telegraph

❑ 1844: First telegraph line (Washington-Baltimore) became operational

❑ Sent electronic signals using wires

❑ Morse devised a language with a series of long and short signals that represented letters and numbers

A	.-	M	--	Y	-.--	6	-....
B	-...	N	-.	Z	--..	7	--...
C	-.-.	O	---	Ä	.-.-	8	---..
D	-..	P	.-.	Ö	---.	9	-----
E	.	Q	--.	Ü	..--	.	.-.-.
F	..-.	R	.-	Ch	----	,	---..
G	--.	S	...	0	-----	?	..-..
H		T	-	1	.-----	!	..---
I	..	U	..-	2	..---	:	-----
J	.----	V	...-	3	...--	"	.-.-.
K	-.-	W	.-	4	-	'	.-....
L	.-..	X	-.-	5		=	---..



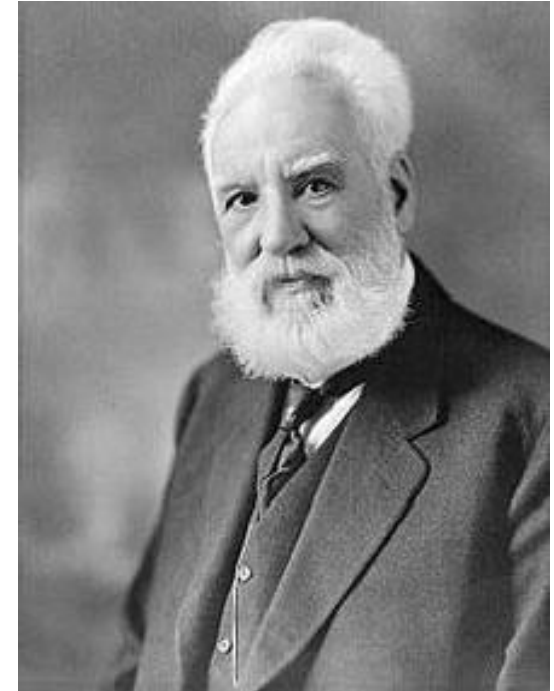
Original Samuel Morse telegraph

History of communication

- ❑ 1854: 25,000 miles of telegraph wires have been laid across the US
 - Train schedules, weather, important news
- ❑ 1864: A telegraph line spans the entire continental US
 - Western Union is formed
 - Very costly

History of communication

- ❑ **1864:** Maxwell formulated the electromagnetic (EM) theory
- ❑ **1875:** Bell invented the telephone
 - Wife was deaf
 - Wanted to communicate to her at a distance
- ❑ **1887:** Hertz demonstrated physical evidence of EM waves
 - Hertz demonstrated the wave character of electrical transmission through space



Alexander Graham Bell
(1847-1922)

History of communication

❑ 1890- 1900: Guglielmo Marconi

- first demonstration of wireless telegraphy
 - Across Atlantic Ocean
 - From Cornwall to Canada
- long wave transmission
- high transmission power necessary (> 200kw)

❑ 1907: Commercial radio broadcast

- huge base stations

❑ 1918: Armstrong invented superheterodyne radio receiver (and FM in 1933)

❑ 1921: land-mobile communication

History of communication

- ❑ 1928: Nyquist sampling theorem
- ❑ 1947: Microwave relay system
- ❑ 1948: Information theory
- ❑ 1957: Era of satellite communication began
- ❑ 1966: Kuen Kao pioneered optical fibre communications (Nobel Prize Winner)
 - Now backbone communication is based on optical fiber
- ❑ 1970's: Era of computer networks began
- ❑ 1981: Analog cellular system (1G)
- ❑ 1988: Digital cellular system launched in Europe
- ❑ 1992: Start of GSM (2G)

History of communication

❑ 1996: HiperLAN (up to 155Mbit/s)

❑ **1997: Wireless LAN - IEEE802.11**

- IEEE standard, 2.4 - 2.5GHz and infrared, 2Mbit/s

❑ 1999: Standardization of additional wireless LANs

- IEEE standard 802.11b, 2.4-2.5GHz, 11Mbit/s
- Bluetooth for pico nets, 2.4Ghz, <1Mbit/s

❑ **2000: 3G network**

❑ **2005: WiMax standardized**

- Not implemented in north America or Europe
- In the recent years, some countries in Asia and Africa implemented WiMax

❑ **2008: 4G LTE**

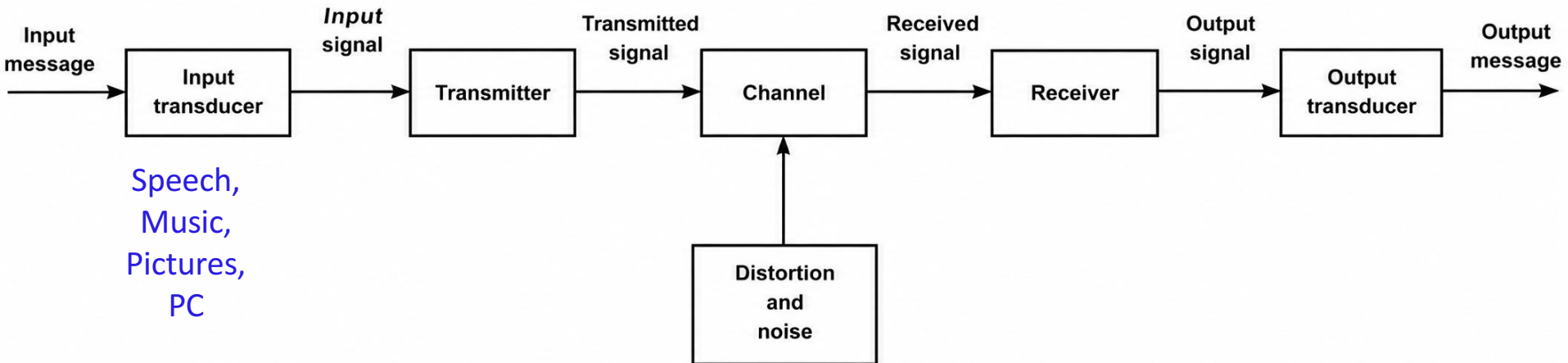
❑ **2018: 5G**

❑ **2030: 6G (expected)**

Top Telecom Manufacturers (Vendors)



Basic Communication System



❑ Communication involves the **transfer of information from one point to another** through a succession of processes

❑ Three basic elements

- **Transmitter:** converts message into a form suitable for transmission
- **Channel:**
 - the physical medium which transports the message signal and delivers it to a receiver
 - introduces **attenuation, distortion, noise, interference**
- **Receiver:** reconstruct a recognizable form of the message

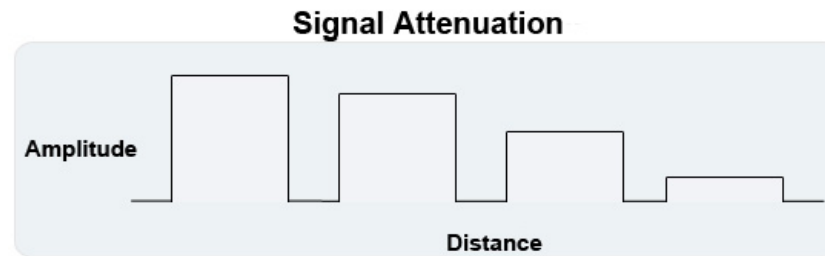
Challenges in Communication

❑ Channel impairments:

Attenuation, Distortion, Noise, Multi-user Interference

- The magnitude of the channel impairments depends on the type of channel

Attenuation:



- Signal attenuation or degradation exists in all media
- Increases with distance
- Wireless medium has the highest attenuation
- Optical fibers have less attenuation (as low as 0.2 dB/km)

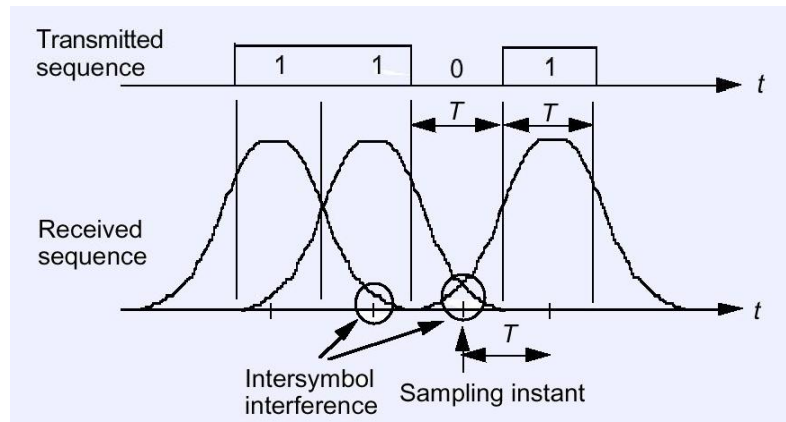
Challenges in Communication

Distortion:

❑ Signals distort during travel through medium (**why?**)

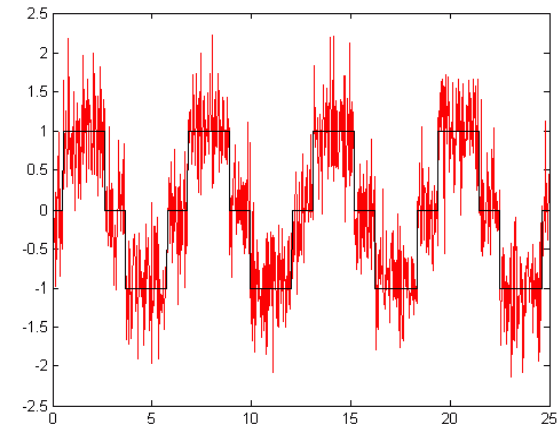
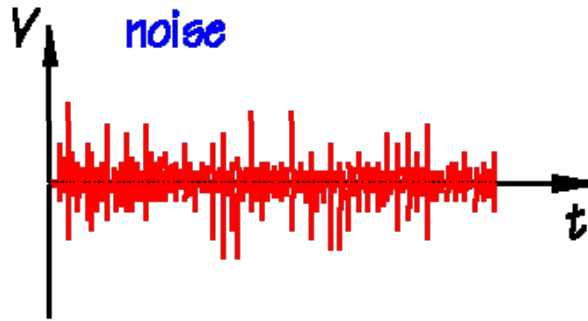
- **Wire:** frequency dependent attenuation
- **Optical fiber:** Delay differences in different modes, frequency dependent attenuation, highest dispersion
- **Wireless:** Delay differences due to multi-path propagation, time dependent randomness of particles, frequency dependent attenuation => highest distortion

❑ Inter-symbol interference due to distortion



Challenges in Communication

Noise:



□ Channel noise/ External noise

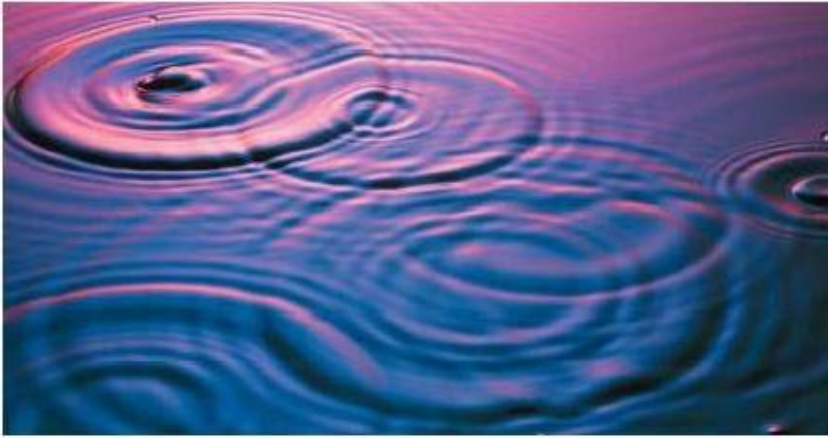
- Random, undesirable electronic energy that enters the communication system via the communicating medium and interferes with the transmitted message
- Human made noise (automobile ignition radiation, microwave oven), natural noise (lightning)
- External noise can be minimized with proper design

□ Receiver background noise/Internal noise

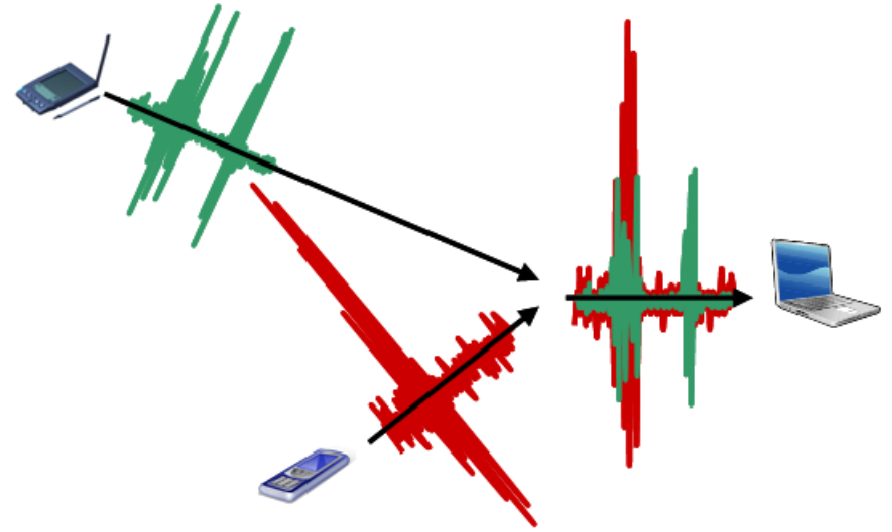
- Thermal noise and random emission in electronic devices
- One of the main problems in communication

Challenges in Communication

Interference: Unwanted signal as like noise, but more structured



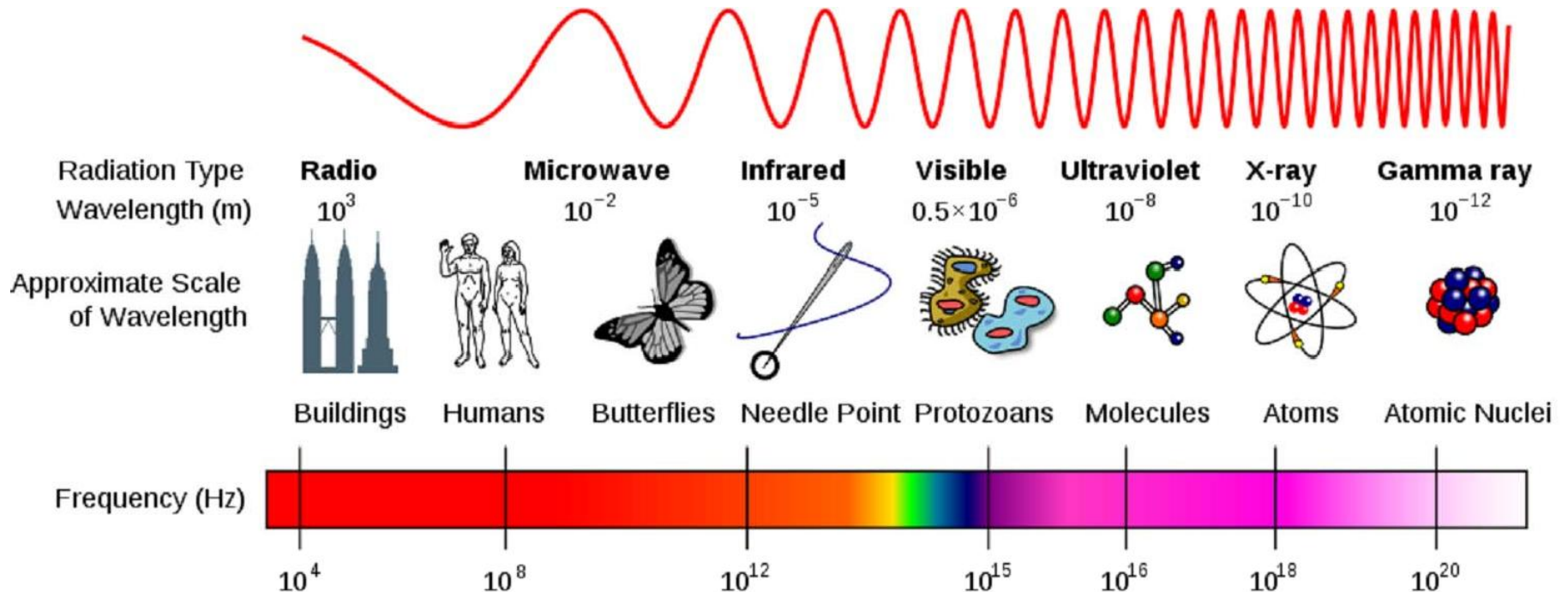
Interference of waves



- Electromagnetic interference (EMI)
- Co-channel interference (CCI)
- Adjacent channel interference (ACI)
- Inter-carrier interference (ICI)

Electromagnetic Spectrum

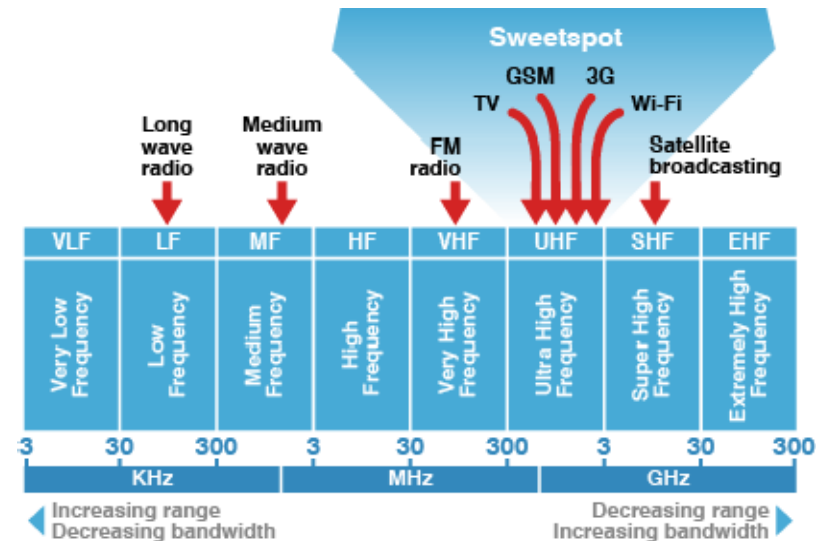
Electromagnetic spectrum is the range of all possible frequencies of **electromagnetic radiation**



Radio Spectrum (3 kHz – 3 THz)

Table of ITU Radio Bands

Band Number	Symbols	Frequency Range	Wavelength Range†
4	VLF	3 to 30 kHz	10 to 100 km
5	LF	30 to 300 kHz	1 to 10 km
6	MF	300 to 3000 kHz	100 to 1000 m
7	HF	3 to 30 MHz	10 to 100 m
8	VHF	30 to 300 MHz	1 to 10 m
9	UHF	300 to 3000 MHz	10 to 100 cm
10	SHF	3 to 30 GHz	1 to 10 cm
11	EHF	30 to 300 GHz	1 to 10 mm
12	THF	300 to 3000 GHz	0.1 to 1 mm



***International Telecommunication Union (ITU):** An UN organization which allocates global radio spectrum and satellite orbits

Bangladesh Telecommunication Regulatory Commission (BTRC) is the authority for frequency allocation in Bangladesh

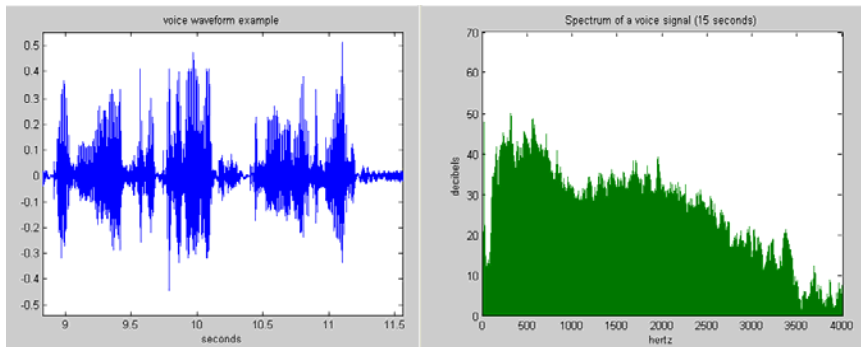
For more details on the application areas:

http://en.wikipedia.org/wiki/Radio_spectrum

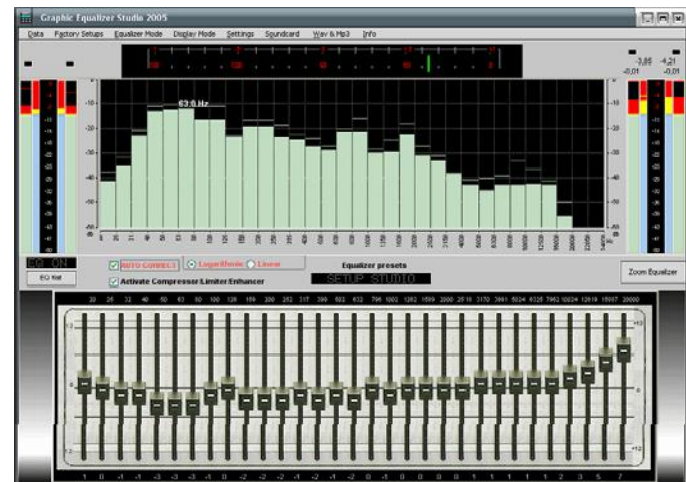
Signal Bandwidth

❑ **Bandwidth:** Measure of the frequency contents of an information signal

❑ Bandwidth of the base band signal – depends on the type of input message



Speech



Music

Channel Bandwidth

❑ **Bandwidth of a communication channel** is the difference between the highest and the lowest frequency that the channel will allow to pass through it



Copper wire: 1 MHz



Coaxial cable: 100 ~ 500 MHz



Microwave/RF: GHz

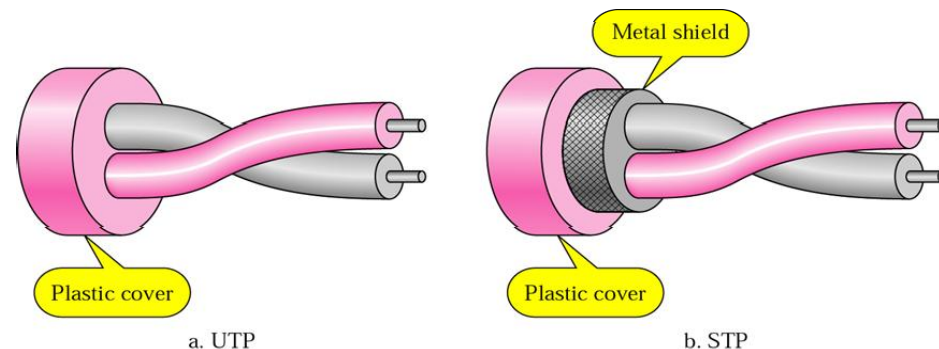
Optical fiber: THz



❑ Bandwidth of a communication channel must be equal or greater than the bandwidth of the information.

Twisted Pair Copper Cable

- Two insulated copper wires arranged in regular spiral pattern
- Number of pairs are bundled together in a cable
- Least expensive and widely used
- Twisting reduces the tendency to radiate radio frequency noise as the radiations from twisted wires cancel each other
- Twisting also decreases the crosstalk/EMI between adjacent pairs in the cable
- Limited in terms of data rate and distance
- Requires amplifiers every 5-6 km for analog signals and repeaters every 2-3 km for digital signals
- Attenuation is a strong function of frequency: Higher frequency implies higher attenuation
- Susceptible to interference and noise



Unshielded twisted pair (UTP):

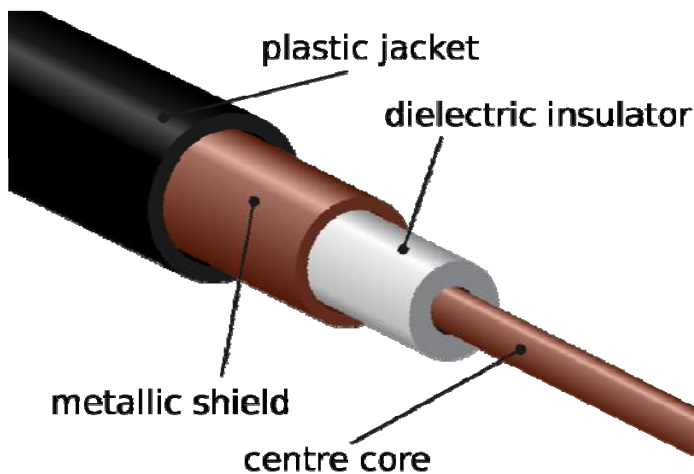
- Speed between 10-100 Mbps
- Susceptible to crosstalk

Shielded twisted pair (STP):

- Supports higher data rate than UTP
- Less susceptible to crosstalk

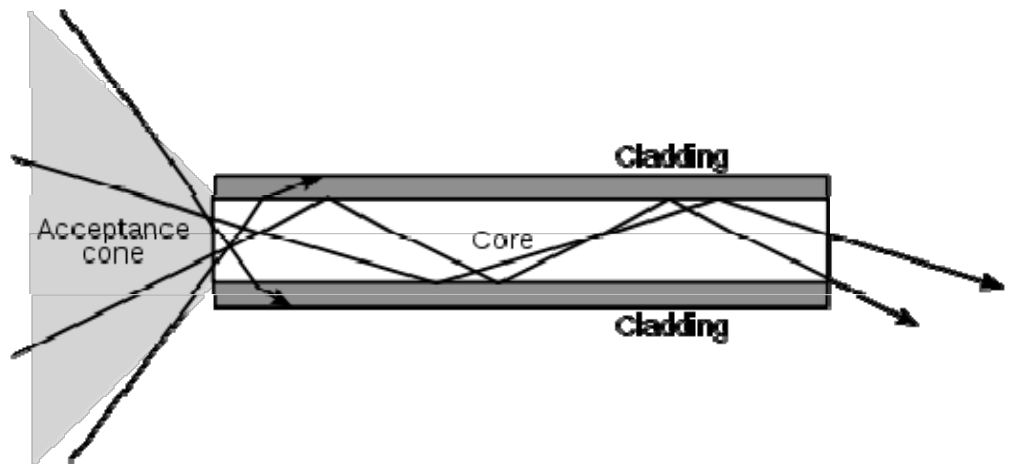
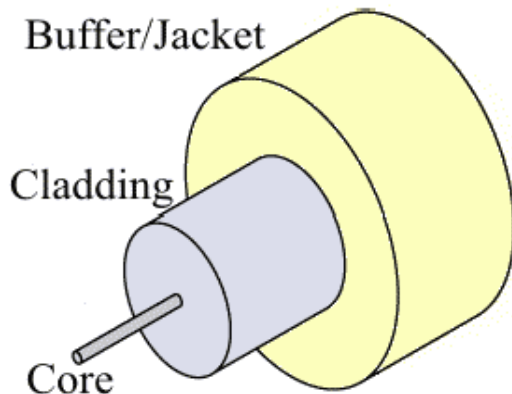
Coaxial Cable

- Contain **two conductors** sharing common axis: a central conductor wire and a surrounding outer conductor/shield serving as ground.
- Most common due to inexpensive, light, flexible and easy to work with.
- Less susceptible to interference and noise as the outer conductor blocks EMI.
- Support higher data rates and longer distance than twisted pair cable.
- Superior frequency characteristics compared to twisted pair.
- Requires amplifiers and repeaters for every few kms.



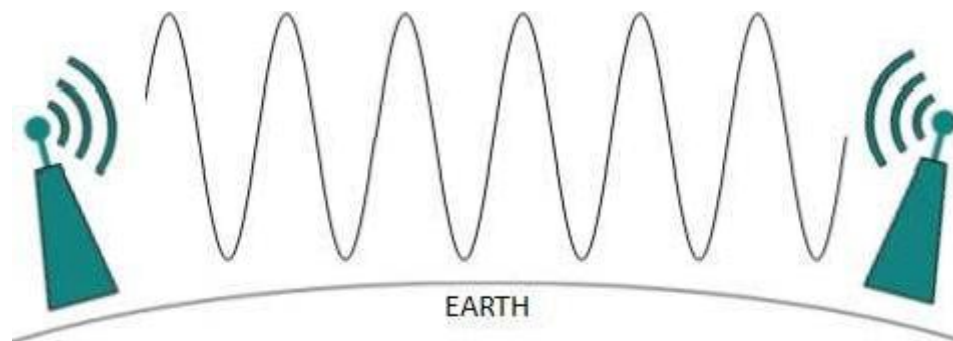
Optical Fiber

- Dielectric waveguide that uses **pulse of light** instead of electrical signals
- Thin and flexible material to guide optical rays
- Cylindrical cross-section with **three concentric links**: Core, Cladding and Jacket
- **Advantages:**
 - Much higher bandwidth (theoretically 2×10^{13} Hz): can carry hundreds of Gbps over tens of kms
 - Smaller size and light weight
 - Significantly lower attenuation (as low as 0.2dB/km): Greater repeater spacing
 - Not affected by external EM fields, i.e., not vulnerable to interference, impulse noise, or crosstalk
 - Ruggedness and flexibility



Wireless Media

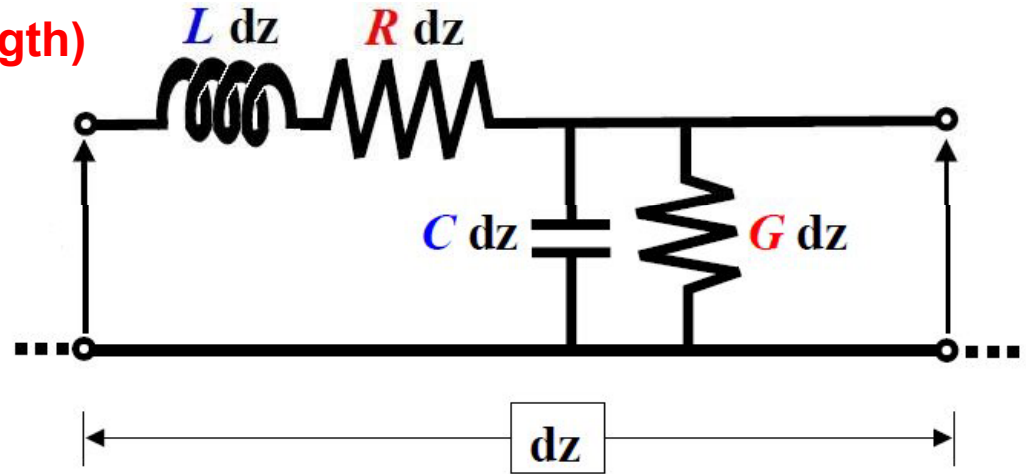
- **No use of wire:** use Earth's atmosphere to act as transmission media
- Transmission and reception are achieved using antenna
- Transmitter sends out the EM signal into the medium
- Receiver picks up the signal from the surrounding medium
- Convenient: Supports mobility and flexibility
- Lower capital and operating expenditure compared to wired networks
- Connection quality vary randomly with time due to fading
- Susceptible to multi-user interference
- Less secured
- Network management more complex



Channel Characteristics

Propagation constant (per unit length)

$$\begin{aligned}\gamma &= \alpha + j\beta = \sqrt{(R + jX)(G + jB)} \\ &= \sqrt{(R + j\omega L)(G + j\omega C)}.\end{aligned}$$



□ Attenuation constant, α : Determines the **attenuation** of a signal of frequency ω over a unit distance given in **nepers/unit length**.

□ Phase constant, β : Determines the **phase change (delay)** in a signal of frequency ω over a unit distance given in **radians/unit length**.

$$\begin{aligned}L &= 10 \log_{10} \frac{x_1^2}{x_2^2} && \text{dB} \\ &= 10 \log_{10} \left(\frac{x_1}{x_2} \right)^2 && \text{dB} \\ &= 20 \log_{10} \frac{x_1}{x_2} && \text{dB} \\ &= \ln \frac{x_1}{x_2} && \text{Np.}\end{aligned}$$

Channel Characteristics

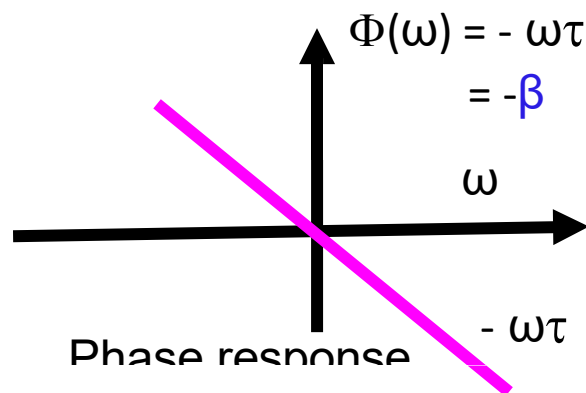
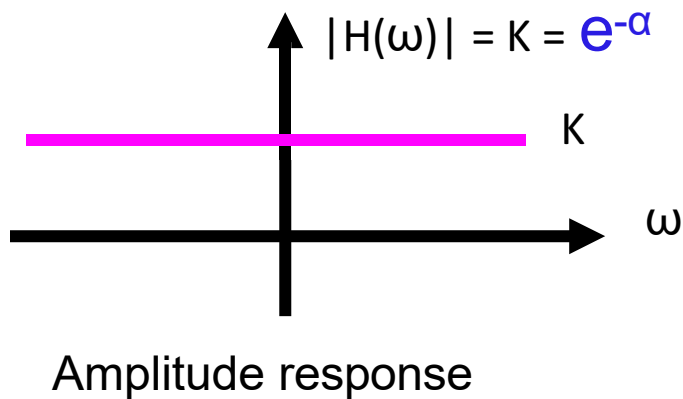
For distortionless transmission:

Transmission is said to be distortion less if the input and output have identical wave shapes: (i) amplitudes of all the frequency components are multiplied by the same factor, and (ii) all the frequency components are delayed by the same amount.

Thus, in distortion-less transmission, the input $x(t)$ and output $y(t)$ satisfy the condition:

$$y(t) = Kx(t - \tau) \Rightarrow Y(\omega) = KX(\omega)e^{-j\omega\tau} = KX(\omega)e^{j\Phi(\omega)}$$

where τ is the delay time and K is a constant.



Phase delay:

$$\tau_p = -\Phi(\omega)/\omega \\ = \beta/\omega$$

Group delay:

$$\tau_g = -d\Phi(\omega)/d\omega \\ = d\beta/d\omega$$

□ τ_p and τ_g both are constant for distortion-less system

➤ Equalizer can be used for minimize amplitude and phase distortions

Channel Capacity

- ❑ Shannon's Capacity Formula (1948):

$$C = B \log_2 (1 + \text{SNR}), \text{ bps}$$

C = capacity (bps), B = channel bandwidth (Hz), $\text{SNR} = \frac{\text{signal power}}{\text{noise power}}$

- ❑ Capacity increases linearly with bandwidth, but only logarithmically with signal strength.
- ❑ **Shannon's limit tells us what can be achieved. But, it tells nothing on how to accomplish it.**

- ❑ **Two primary resources in communications:**

- Transmitted power (should be green, i.e., lower energy requirement)
- Channel bandwidth (very expensive)

National Frequency Allocation Plan (NFAP) prepared by BTRC describes which frequency to be used for which applications in Bangladesh

